Robust Land Cover Classification With Multimodal Knowledge Distillation

Guozheng Xu, Xue Jiang, *Senior Member, IEEE*, Yue Zhou, *Student Member, IEEE*, Shutao Li, *Fellow, IEEE*, Xingzhao Liu, *Member, IEEE*, and Peiwen Lin

***Abstract*— In recent years, enormous studies have been conducted to improve the land cover (LC) classification performance of multimodal remote sensing (RS) data, which outperforms single-modal-based methods by a large margin due to information diversity. To go a step further, we develop a two-branch patch-based convolutional neural network (CNN) with an encoder–decoder (ED) module to fuse multimodal RS data information. A knowledge distillation in model (DIM) module is proposed to guild per-modality encoder learning with the final fused information to enable multimodal data fusion more effectively. Moreover, utilizing multimodal information to guide single-modal learning still remains to be explored. To this end, a knowledge distillation cross-model (DCM) module is designed to improve single-modal LC classification with multimodal knowledge distillation, which bridges the gap between single-modal-based and multimodal-based methods. In particular, the multimodal-based method is taken as a teacher to transfer knowledge to single-modal-based methods. Extensive experiments are carried out on two multimodal RS datasets, including hyperspectral (HS) and light detection and ranging (LiDAR) data, i.e., the Houston2013 dataset, and HS and synthetic aperture radar (SAR) data, i.e., the Berlin dataset. The results demonstrate the effectiveness and superiority of the proposed multimodal fusion strategy in comparison with several state-of-the-art multimodal RS data classification methods. Also, the proposed DCM module improves the LC classification performance of single-modal methods by a large margin.**

***Index Terms*— Knowledge distillation, land cover (LC) classification, multimodal, remote sensing (RS) data, single-modal.**

I. Introduction

LAND cover (LC) classification is one of the most important tasks in Earth observation research, which provides basic information for high-level image interpretation and analysis of remote sensing (RS) [1], [2], [3], [4]. Driven by the rapid development of RS devices and platforms, various modalities of RS data become available, which provides more exploration opportunities for the researchers to demonstrate the benefit of combining such sensors' information [5], [6], [7]. Remote sensors can be categorized into two classes: passive ones, e.g., hyperspectral (HS) and multispectral (MS) and active ones, e.g., synthetic aperture radar (SAR) and light detection and ranging (LiDAR). These sensors enable us to classify LC with various complementary information of the same object. For example, HS and MS can provide detailed spectral and spatial information while LiDAR and SAR can record the elevation information of the objects. On the other hand, passive sensors relying on natural illumination are limited to environmental conditions, while active sensors relying on their own illumination sources operate well during the night and in shaded areas. To this end, developing an advanced multimodal data processing and analysis approach has gained significant attention recently in the RS community [8], [9], [10], [14].

To address this problem, some traditional multimodal algorithms have been proposed, such as morphological profiles (MPs) [15], [16], [17] and subspace learning-based approaches [18], [19], [20], [21]. These conventional methods require expertise and prior knowledge to design appropriate feature extraction and feature selection methods. The poor capability of handcrafted features also limits the performance of these methods. Moreover, handcrafted features are not suitable to close the large gap between different modalities [8], [9], [22], [23].

In recent years, various deep learning (DL) networks, such as convolutional neural networks (CNNs) [24], recurrent neural networks (RNNs) [25], and graph convolutional networks (GCNs) [26], can learn high-level semantic features from data itself in an end-to-end manner and have shown great advantages against handcrafted features. Some DL-based multimodal networks have been proposed for LC classification tasks to break information bottlenecks and outperform the methods that use a single modality by a large margin. For monosource satellite data, researchers [27], [28], [29] fused multitemporal spatial data to improve the classification performance. Li et al. [12] presented a transfer learning approach using a multimodel feature extraction framework for high-resolution RS image classification, achieving high accuracy and exploring its application to HS data. Liu and Qin [11] proposed a domain adaptation method and transfer learning framework for multimodal RS data, reducing dataset discrepancies and potentially reducing label costs in classification. Progressive domain adaptation (PDA) [13] using a hybrid multimodel framework is proposed for

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Guozheng Xu, Xue Jiang, Yue Zhou, and Xingzhao Liu are with the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: guozhengxu@sjtu.edu.cn; xuejiang@sjtu.edu.cn; sjtu_zy@sjtu.edu.cn; xzhliu@sjtu.edu.cn).

Shutao Li is with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China (e-mail: shutao_li@hnu.edu.cn).

Peiwen Lin is with SenseTime, Shanghai 200030, China (e-mail: peiwenlin@tju.edu.cn).

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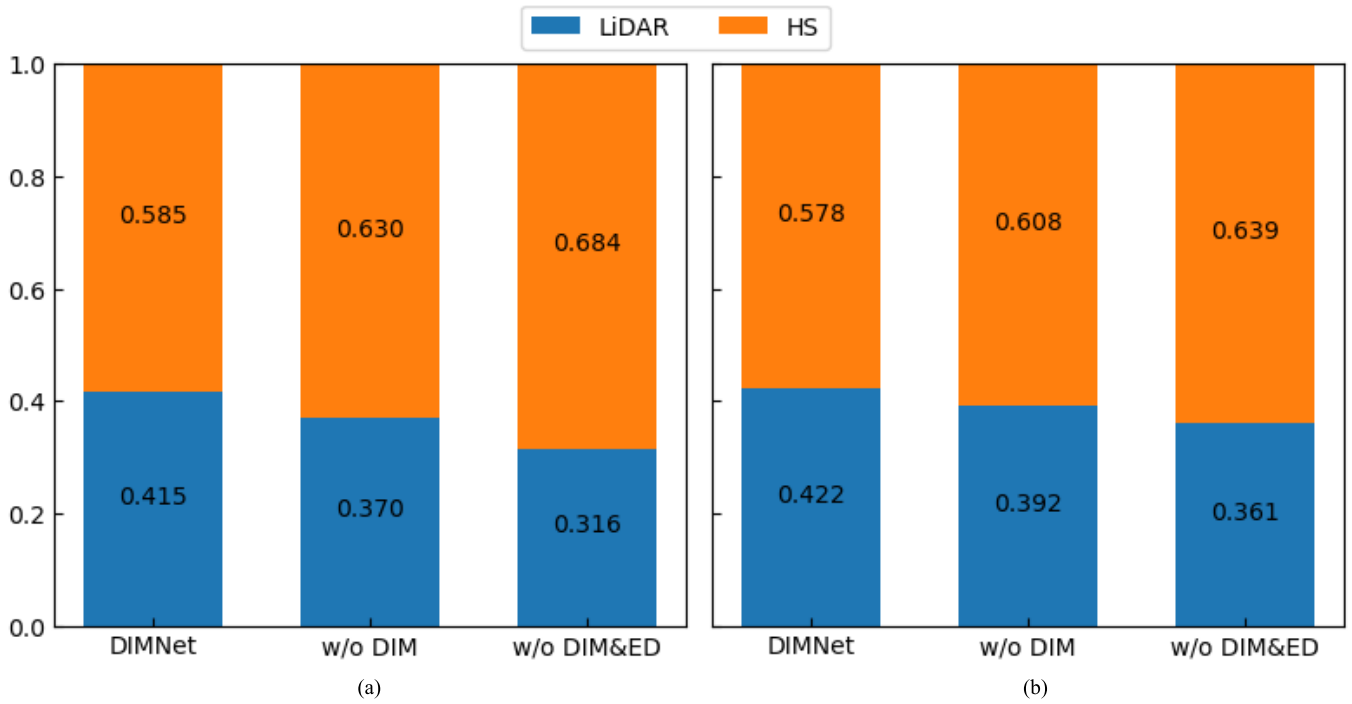


Fig. 1. Visualization example of the impact of different modules on neuron activation across the heterogeneous HS and LiDAR data in the fusion layer. (a) Number of activated neurons. In detail, we classify neurons with weights exceeding the mean as activated neurons. We conducted separate quantifications of the activated neurons originating from the HS and LiDAR modalities and calculated their respective proportions. (b) Sum of neuron weights. In detail, we conducted separate computations of the summation of neuronal weights originating from the HS and LiDAR modalities and calculated their respective proportions.

change detection in season-varying RS images, effectively suppressing seasonal variation-induced changes and emphasizing natural and human activity-caused changes. For highly heterogeneous data, Chen et al. [24] first investigated the potential of a two-stream CNNs model for RS data fusion. Similar to [24], Co-CNN [9] also utilized two CNNs to extract features and integrated the HS and LiDAR data in an advanced cascade fashion via a parameter-sharing strategy. Hong et al. [8] focused on "what," "where," and "how" to fuse multimodal RS data and studied five fusion architectures, including early, middle, late, encoder–decoder (ED), and cross fusion. Furthermore, CCR-Net [30] designed a cross-channel reconstruction mechanism to better eliminate the gap between different modalities and obtained diagnostic features.

These recent approaches have shown effectiveness in handling multimodal data, yet the existing methods still do not make full use of multimodal information. A prevalent issue observed in numerous existing models, as depicted in Fig. 1 with the cases of "w/o DIM" and "w/o DIM&ED," is that the fusion layer's information tends to be dominated by a single modality. This tendency can inadvertently skew the model performance, underscoring the need for a more balanced integration of modalities. Due to the nonconvex nature of neural networks, it is common to encounter local optima during the training process instead of reaching the global optimum. In the case of optical RS images, which usually contain richer information, the fusion layer tends to rely more on this modality during training. As a result, the training loss decreases at a faster rate, leading to higher activation of neurons associated with the optical RS images. However, when the network becomes trapped in such a local optimum, it tends to underutilize information from the other modality.

Moreover, the utilization of multimodal data to enhance the learning of single-modal models is seldom considered. During training, the data is prepared in advance and is controllable, so we can remove samples with missing modalities. However, when applying the model to real-world production, we cannot guarantee that every sensor could work properly at all times. For instance, optical sensors may fail to capture clear images due to cloud cover, or certain sensors might be temporarily out of service due to damage. To address this problem, Hong et al. [8] directly zeroed one of the modalities during the inference stage and found compactness-based fusion strategy was better to handle this situation. However, multimodal information does not explicitly aid single-modal learning.

Distillation learning has recently attracted more attention, which concentrates on transferring knowledge from a heavy model to a light one to improve the light model's performance without introducing extra costs. Motivated by the distillation approaches [31], [32], [33], in this article, we strive to develop a simple but effective framework via distillation modules to better mine multimodal information. More in detail, we first train a multimodal model, named DIMNet, which is a two-branch patch-based CNN framework with an ED module to obtain more compact feature representations. Then, we design a knowledge distillation in model (DIM) module, which enables the network architecture to distill the knowledge from the fused feature to the per-source feature with the aim of improving the final multimodal classification performance. Second, we train single-modal models with the

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DIMNet. In this stage, we only optimize the single-modal model, while the DIMNet is well-trained and the weights are fixed. We propose a knowledge distillation cross-model (DCM) module, which transfers the multimodal information to single-modal models to improve the single-modal classification performance. Since multimodal models and single-modal models are independent, we can infer the single-modal models independently. The major contributions of this article are summarized as follows.

1) We develop a novel two-branch patch-based CNN framework with multimodal RS data, called DIMNet. Well-designed ED and DIM modules in DIMNet are presented to fuse highly heterogeneous modalities in a more effective way.
2) DCM module is proposed to help single-modal model learning with multimodal RS data during the training stage. We utilize DIMNet (teacher model) to guide the optimization direction of the single-modal model (student model), which improves the classification performance with only a certain modality available during the inference stage without introducing extra costs.
3) Beyond substantial experiments validation of the proposed approach, we have also provided comprehensive theoretical and visual elucidations of the modules we have introduced to delineate their functionalities. DIMNet enables the avoidance of a single modality dominating the fusion layer. The DCM module facilitates the injection of class relationship prior knowledge from a multimodal model to a single-modal model. These findings could potentially shed light on future research in the domain of multimodal fusion.

II. Methodology

Different from the existing CNN models, the proposed framework is trained in two stages. In the first stage, we train a multimodal fusion model (named DIMNet) with ED and DIM modules. In the second stage, we train single-modal models with a DCM module.

A. Feature Extraction via Patch-Based CNN

In recent years, CNN has been widely used to extract discriminative features in LC classification task [9], [10], [24], [30]. In this article, we develop a two-branch patch-based CNN feature extractor used for each modality. The structure of the patch-based CNN feature extractor is the same for each modality except for the first convolution layer due to the fact that the different modalities have different numbers of channels. Pixels in a neighborhood region of a given pixel in the original data are taken into consideration. In detail, given two modalities data $\boldsymbol{X}_1 \in \mathbb{R}^{h\times w\times d1}$ and $\boldsymbol{X}_2 \in \mathbb{R}^{h\times w\times d2}$ covering the same area. Here, h and w are the height and width, respectively. d_1 and d_2 are the numbers of channels of the two modalities, respectively. For each pixel, the patches $\boldsymbol{x}_1 \in \mathbb{R}^{k\times k\times d1}$ and $\boldsymbol{x}_2 \in \mathbb{R}^{k\times k\times d2}$ centered at it are chosen from $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$, respectively. For a fair comparison, we set the neighboring size k to 7 as in [30]. Then, $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are fed into CNN feature extractor. The lth layer output $\boldsymbol{u}_s^{(l)}$ can be formulated as

$$\boldsymbol{u}_{s,i}^{(l)} = \begin{cases} f\left(\boldsymbol{W}_s^l \boldsymbol{x}_{s,i} + \boldsymbol{b}_s^{(l)}\right), & l = 1 \\ f\left(\boldsymbol{W}_s^l \boldsymbol{z}_{s,i}^{l-1} + \boldsymbol{b}_s^{(l)}\right), & l = 2, \ldots, n \end{cases} \tag{1}$$

where $\boldsymbol{x}_{s,i}$, $s = 1, 2$ denote the ith pixel in the sth modality and n is the number of layers in CNN. $z_{s,i}^{l-1}$ denote the intermediate variables in the second-nth layers. f is the activate function and is set as rectified linear unit (ReLU). $\boldsymbol{W}_s^l$ and $\boldsymbol{b}_s^{(l)}$ are weights and biases.

The patch-based CNN feature extractor for each modality stacks two CNN blocks, which consist of a 3×3 convolutional layer, batch normalization [35] layer, 2×2 max-pooling layer, and ReLU activation layer.

B. Feature Fusion

To fuse two modalities of information, the concatenation-based fusion method is widely used [9], [24]. However, this method often results in the fact that the neurons from one modality are activated and those from another modality are inhibited, especially for heterogeneous data [8], [30]. Alternatively, fusing multimodal features in a more compact way is a feasible solution [8], [30], [36]. An ED module is designed to yield more compact feature representations via a self-supervised learning strategy. In detail, we reconstruct the features of each modality from the fused features. The encoder part can be formulated as

$$\boldsymbol{v}_i = g_{\text{en}}([\boldsymbol{u}_{1,i}, \boldsymbol{u}_{2,i}], \boldsymbol{W}_{\text{en}}, \boldsymbol{b}_{\text{en}}) \tag{2}$$

where $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ are the features extracted from each modality. $\boldsymbol{W}_{\text{en}}$ and $\boldsymbol{b}_{\text{en}}$ are the weights and biases in the encoder. g_{en} is the encoder network and we set it as a fully connected (FC) layer. The decoder part can be formulated as

$$\boldsymbol{z}_i = g_{\text{de}}(\boldsymbol{v}_i, \boldsymbol{W}_{\text{de}}, \boldsymbol{b}_{\text{de}}) \tag{3}$$

where $\boldsymbol{W}_{\text{en}}$ and $\boldsymbol{b}_{\text{en}}$ are the weights and biases in the encoder. g_{de} is the decoder network and we set it as an FC layer. $\boldsymbol{z}$ can also be written as $[\boldsymbol{z}_1, \boldsymbol{z}_2]$. Therefore, the consistency loss can be formulated as

$$L_{\text{cons}} = \sum_{i=1}^{N} \left| [\boldsymbol{u}_{1,i}, \boldsymbol{u}_{2,i}] - [\boldsymbol{z}_{1,i}, \boldsymbol{z}_{2,i}] \right|_2^2. \tag{4}$$

C. Knowledge Distillation

Inspired by the success of the distillation method that can transfer knowledge in a teacher–student paradigm to help a small student model learn effectively from a large teacher model for visual intelligence [37], [38], [39], we introduced the distillation method in our framework. Different from the previous studies with the purpose of improving the learning ability of small models, we leverage a distillation strategy that supports the network to transfer knowledge in model and cross-model to improve both multimodal and single-modal learning better.

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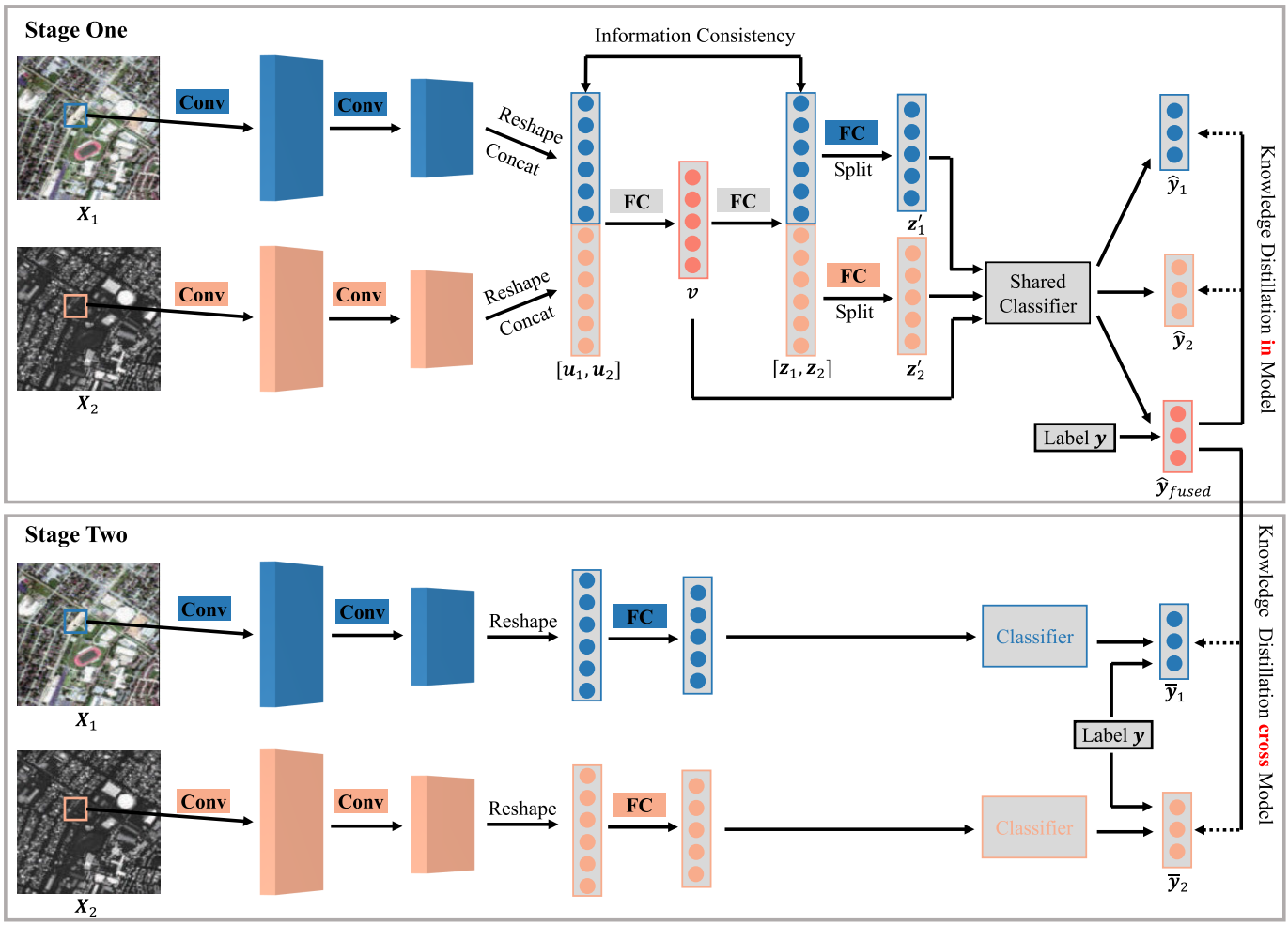


Fig. 2. Illustration of our proposed framework. "Conv" block consists of a 3 × 3 convolutional layer, batch normalization layer, 2 × 2 max-pooling layer and ReLU activation layer. "FC" block consists of a fully connected layer, batch normalization layer, and ReLU activation layer. "Shared Classifier" and "Classifier" have the same structure and consist of "FC" blocks and a softmax layer.

1) Knowledge DIM: The DIM module leverages a self-distillation component that supports our network to learn from itself. The per-source output layer is trained to mime the behavior of the final fused output with the goal of distilling knowledge from fused information to the per-source stream. In detail, to align the dimension $\boldsymbol{v}$, $\boldsymbol{z}_1$, and $\boldsymbol{z}_2$, a FC layer is applied to $\boldsymbol{z}_1$ and $\boldsymbol{z}_2$

$$\boldsymbol{z}'_{s,i} = f(\boldsymbol{W}_s \boldsymbol{z}_{s,i} + \boldsymbol{b}_s). \tag{5}$$

Then, $\boldsymbol{v}$, $\boldsymbol{z}'_1$, and $\boldsymbol{z}'_2$ are fed into a shared classifier to obtain the probability of each category, which consists of FC layers and a softmax layer. As shown in Fig. 2, the supervision signal of the category is only applied to the probability of fused output

$$L_{\text{cls}} = -\frac{1}{N}\sum_{k=1}^{N}\left[\boldsymbol{y}^k \log\left(\hat{\boldsymbol{y}}^k_{\text{fused}}\right) + \left(1-\boldsymbol{y}^k\right)\log\left(1-\hat{\boldsymbol{y}}^k_{\text{fused}}\right)\right] \tag{6}$$

where N represents the number of training samples. $\boldsymbol{y}^k$ is the ground-truth for the kth sample. And the output probability $\hat{\boldsymbol{y}}^k_{\text{fused}}$ of the shared classifier is the classification result of $\boldsymbol{v}$ for the kth sample. Then, the self-distillation strategy in the model can be formalized as

$$L_{\text{DIM}} = -\frac{1}{N}\sum_{k=1}^{N}\sum_{s=1}^{2}\left[\hat{\boldsymbol{y}}^k_{\text{fused}} \log\left(\hat{\boldsymbol{y}}^k_s\right) + \left(1-\hat{\boldsymbol{y}}^k_{\text{fused}}\right)\log\left(1-\hat{\boldsymbol{y}}^k_s\right)\right] \tag{7}$$

where $\hat{\boldsymbol{y}}^k_s$ is the output probability of the sth modality for the kth sample. Also, as in [40], a temperature hyper-parameter T is introduced to adjust the entropy of the predicted softmax-probability distributions before they are used in the loss computation: $\hat{\boldsymbol{y}}_s \propto \exp(\log \hat{\boldsymbol{y}}_s / T)$ and $\hat{\boldsymbol{y}}_{\text{fused}} \propto \exp(\log \hat{\boldsymbol{y}}_{\text{fused}} / T)$.

To this end, in the first stage, the network parameters can be optimized by the following loss function:

$$L_{\text{stage1}} = L_{\text{cls}} + \alpha L_{\text{cons}} + \beta L_{\text{DIM}} \tag{8}$$

where α and β are the hyper-parameters to balance different terms in (8). Considering that the performance of our model is not sensitive to these parameters, α and β are experimentally and empirically set to be 0.1 and 1, respectively.

2) Knowledge DCM: The DCM module leverages a distillation component that transfers the knowledge of the multimodal model to the single-modal models in a teacher–student paradigm. The teacher model is trained in the first stage and

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TABLE I
NUMBER OF TRAINING SAMPLES AND TESTING SAMPLES FOR EACH CLASS IN HS-LIDAR HOUSTON2013 DATASET

No.	Class Name	Training Set	Testing Set
Class 1	Healthy Grass	198	1053
Class 2	Stressed Grass	190	1064
Class 3	Synthetic Grass	192	505
Class 4	Tree	188	1056
Class 5	Soil	186	1056
Class 6	Water	182	143
Class 7	Residential	196	1072
Class 8	Commercial	191	1053
Class 9	Road	193	1059
Class 10	Highway	191	1036
Class 11	Railway	181	1054
Class 12	Parking Lot1	192	1041
Class 13	Parking Lot2	184	285
Class 14	Tennis Court	181	247
Class 15	Running Track	187	473
	Total	2832	12197

fuses the multimodal information, which is clarified above. In order to keep consistent with the teacher model, the student model is also composed of a feature extractor and classifier which is similar to the teacher model. In fact, removing a modal branch from DIMNet and removing ED and DIM modules is a standard single-modal model. In the second stage, the parameters of the teacher model are fixed and only the parameters of the student model are optimized. The supervision signal of the category is applied to the per-source model

$$L_{\mathrm{cls},s} = -\frac{1}{N}\sum_{k=1}^{N}\left[\boldsymbol{y}^k \log\left(\overline{\boldsymbol{y}}_s^k\right) + \left(1-\boldsymbol{y}^k\right)\log\left(1-\overline{\boldsymbol{y}}_s^k\right)\right] \quad (9)$$

where $\overline{\boldsymbol{y}}_s^k$ is the output probability of the sth modality model for the kth sample. Similar to L_{DIM}, the distillation strategy cross model can be formulized as

$$L_{\mathrm{DCM},s} = -\frac{1}{N}\sum_{k=1}^{N}\left[\hat{\boldsymbol{y}}_{\mathrm{fused}}^k \log\left(\overline{\boldsymbol{y}}_s^k\right) + \left(1-\hat{\boldsymbol{y}}_{\mathrm{fused}}^k\right)\log\left(1-\overline{\boldsymbol{y}}_s^k\right)\right]. \quad (10)$$

To this end, in the second stage, the student network parameters can be optimized by the following loss function:

$$L_{\mathrm{stage2},s} = L_{\mathrm{cls},s} + \gamma L_{\mathrm{DCM},s} \quad (11)$$

where γ is the hyper-parameters to balance different terms in (11). Considering that the performance of our model is not sensitive to γ, it is experimentally and empirically set as 1.

III. EXPERIMENTAL RESULTS

A. Datasets Description

We validate the effectiveness of our proposed framework on two popular and widely used multimodal fusion datasets in LC classification. The two datasets are described as follows.

1) *HS-LiDAR Houston2013 Data*[1] *[34]:* The dataset consists of an HS image and LiDAR data with dimension 349 × 1905, which was acquired by the ITRES CASI-1500 imaging sensor over the campus of the University of Houston and the neighboring rural areas in Texas, USA. The HS image consists of 144 spectral channels ranging from 364 to 1046 nm with a 10 nm spectral interval, and the LiDAR has only 1 band. There exist 15 LC categories in the studied scene, and the specific numbers of samples for each category are listed in Table I.

[1]http://www.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

TABLE II
NUMBER OF TRAINING SAMPLES AND TESTING SAMPLES FOR EACH CLASS IN HS-SAR BERLIN DATASET

No.	Class Name	Training Set	Testing Set
Class 1	Forest	443	54511
Class 2	Residential Area	423	268219
Class 3	Industrial Area	499	19067
Class 4	Low Plants	376	58906
Class 5	Soil	331	17095
Class 6	Allotment	280	13025
Class 7	Commercial Area	298	24526
Class 8	Water	170	6502
	Total	2820	461851

2) *HS-SAR Berlin Data*[2] *[41]:* The dataset consists of an EnMAP HS image simulated from the HyMap HS data and Sentinel-1 SAR data with dimension 797 × 220, which covered the Berlin urban and its surrounding rural areas. The HS image consists of 244 spectral bands ranging from 400 to 2500 nm, and the SAR data consists of four bands (VV-VH) from a dual-polarSAR. Moreover, the ground-truth label is obtained from the OpenStreetMap data. There exist 8 LC categories, and the specific numbers of samples for each category are listed in Table II.

B. Evaluation Metrics

Following previous works [8], [14], [30], we use overall accuracy (OA), average accuracy (AA), and kappa coefficient (kappa) as evaluation metrics. In general, higher values of these three indices mean better performance of the model. They can be calculated as follows:

$$\mathrm{OA} = \frac{N_c}{N_a} \quad (12)$$

$$\mathrm{AA} = \frac{1}{C}\sum_{C}^{i=1}\frac{N_c^i}{N_a^i} \quad (13)$$

$$\mathrm{kappa} = \frac{\mathrm{OA}-P_e}{1-P_e} \quad (14)$$

where N_c and N_a are the number of samples classified correctly and the number of total samples, respectively. The superscript i corresponds to the ith class. C is the total number of LC categories. P_e is the hypothetical probability of chance agreement, which can be formulated by

$$P_e = \frac{N_r^1 \times N_p^1 + \cdots + N_r^i \times N_p^i + \cdots + N_r^C \times N_p^C}{N_a \times N_a} \quad (15)$$

[2]http://www.grss-ieee.org/2017-ieee-grss-data-fusion-contest/

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TABLE III
COMPARISON OF DIFFERENT METHODS ON THE HS-LIDAR HOUSTON2013 DATASET. THE BOLD ITEMS DENOTE THE OPTIMAL VALUES IN THE ROWS

Methods	HS-NN	HS-RF	LiDAR-NN	LiDAR-RF	Stack-NN	Stack-RF	CoSpace	LeMA	CapsNet	Co-CNN	CCR-Net	DIMNet
OA (%)	73.89	73.12	10.04	31.49	79.57	76.19	82.14	85.42	86.52	87.23	88.15	**91.47**
AA (%)	76.89	77.04	7.74	37.18	81.81	79.64	84.54	87.05	87.66	88.82	89.82	**92.48**
k (%)	71.92	71.11	3.20	26.36	77.93	74.40	80.62	84.17	85.41	86.19	87.19	**90.77**
Class1	82.24	82.62	**92.12**	13.49	82.62	82.53	81.96	81.86	81.10	83.10	83.00	83.00
Class2	81.77	83.55	21.43	16.26	81.77	83.36	83.27	83.08	81.02	**84.87**	**84.87**	84.68
Class3	99.60	97.62	0.00	56.63	**100.00**	98.42	**100.00**	**100.00**	96.44	99.80	**100.00**	99.01
Class4	91.67	91 .95	0.38	42.80	91.48	91.48	94.22	**94.79**	88.35	92.42	92.14	91.38
Class5	97.06	96.78	0.00	58.05	96.50	96.88	99.34	99.34	**100.00**	99.24	99.81	99.62
Class6	95.10	99.30	0.00	58.04	95.10	98.60	**99.30**	**99.30**	95.80	95.80	95.80	95.10
Class7	73.41	74.72	1.96	41.42	77.99	80.04	81 .44	88.99	86.38	95.24	**95.34**	92.91
Class8	36.94	33.43	0.00	27.64	66.48	43.02	66.10	74.26	**90.88**	81 .86	81.39	87.27
Class9	66.76	70.16	0.09	14.35	70.07	69.97	69.97	73.84	82.53	85.08	84.14	**88.01**
Class10	54.05	43.24	0.100	9.07	58.01	45.85	48.94	72.20	72.78	61.10	63.22	**93.82**
Class11	70.30	69.83	0.00	39.94	74.48	74.38	88.61	82.26	88.99	83.87	**90.32**	88.80
Class12	58.89	54.85	0.00	8.55	76.66	67.82	88.57	90.30	83.09	91.26	93.08	**96.54**
Class13	49.47	60.00	0.00	15.44	57.54	64.56	68.07	67.37	76.14	88.77	88.42	**90.53**
Class14	97.98	99.60	0.00	80.16	100.00	100.00	100.00	100.00	93.93	91.09	96.36	**96.76**
Class15	98.10	97.89	0.00	75.90	98.52	97.67	98.31	98.10	97.46	98.73	99.37	**99.79**

TABLE IV
COMPARISON OF DIFFERENT METHODS ON THE HS-SAR BERLIN DATASET. THE BOLD ITEMS DENOTE THE OPTIMAL VALUES IN THE ROWS

Methods	HS-NN	HS-RF	SAR-NN	SAR-RF	Stack-NN	Stack-RF	CoSpace	LeMA	CapsNet	Co-CNN	CCR-Net	DIMNet
OA (%)	50.30	54.94	21.98	32.86	51.71	55.82	64.53	66.71	66.55	68.51	69.85	**73.73**
AA (%)	58.39	58.47	21.18	34.65	59.04	58.50	62.51	62.35	61.18	62.34	**66.99**	66.60
K (%)	37.09	41.34	7.59	17.90	38.38	42.08	50.93	53.12	52.77	54.76	57.16	**61.11**
Class1	71.45	79.29	24.47	32.53	73.57	79.31	85.09	85.11	84.96	84.09	**85.93**	74.46
Class2	41.19	48.68	19.86	30.72	42.93	50.46	61.60	64.84	65.22	68.48	68.07	**76.90**
Class3	40.37	40.84	25.53	36.60	38.72	41.64	51.18	48.94	48.42	49.09	**53.17**	48.56
Class4	68.07	66.68	31.79	44.07	69.23	65.82	75.44	80.04	80.80	79.43	82.62	**85.28**
Class5	81.05	67.15	31.61	52.00	81.01	66.94	82.50	80.66	69.18	81.25	85.10	**86.11**
Class6	**63.89**	61.19	13.32	33.37	62.89	61.01	54.66	54.07	55.08	50.68	63.02	56.70
Class7	35.31	**37.01**	14.96	13.04	36.83	35.84	28.81	27.40	26.12	26.16	29.23	29.18
Class8	65.76	66.90	7.91	34.88	67.13	66.98	60.78	57.75	59.69	59.52	68.78	**75.64**

where N_r^i and N_p^i are the numbers of real samples and predicted samples for the i class.

C. Experimental Settings

The proposed framework is implemented on the Tensorflow programming platform. Following CCR-Net [30], the size of the input patches in networks is 7×7 pixels. The Adam [42] is adopted as the optimizer to update the network gradient with the exponential learning rate policy, where the initialized learning rate and power are set to 0.001 and 0.5, respectively. The parameter of networks is initialized with He initialization [43]. In addition, the momentum is parameterized by 0.9 and the batch size is set to 64.

D. Results and Analysis

To demonstrate the effectiveness of the proposed self-distillation strategy for fusing multimodal information, several representative algorithms are selected for both quantitative and qualitative comparisons, including single-modal algorithms, stacked feature algorithms, CoSpace [20], LeMA [21], CapsNet [34], Co-CNN [9], CCR-Net [30], and ours. It is worth noting that the proposed algorithm mainly aims to show the effectiveness and superiority of the distillation module in fusing multimodal information rather than investigate the hyperparameter setting in networks (e.g., four or five layers in feature extraction or bigger patch size will result in better performance, please see [8], [9], [30]). These algorithms mentioned above are designed as consistently as possible for a fair comparison followed by CCR-Net [30].

1) Comparison of the Multimodal Results: Tables III and IV show the quantitative comparison between the proposed DIMNet method and other algorithms in terms of three commonly used indices OA, AA, and kappa as well as the accuracy for each category.

Due to the rich spectral information in HS modality, the networks using HS modality perform dramatically better than those using LiDAR or SAR with both NN and RF classifiers. As expected, the networks using single-modal input, e.g., HS single and LiDAR single, do not perform well compared to that of the complementary multimodal inputs, e.g., HS and LiDAR in Houston2013 Data. CoSpace [20] and LeMA [21] are two excellent traditional algorithms via aligning multimodal training data. Moreover, recent deep-learning-based methods (CapsNet [34], Co-CNN [9], CCR-Net [30], and ours) have more robust feature extraction capability and the ability to

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TABLE V

COMPARISON OF SINGLE-MODAL RESULTS WITH AND WITHOUT DISTILLATION ON THE HS-LIDAR HOUSTON2013 DATASET. THE BOLD ITEMS DENOTE THE OPTIMAL VALUES IN THE ROWS

Methods	HS_Single	HS_DCM	LiDAR_Single	LiDAR_DCM
$OA(\%)$	80.79	**82.75**	68.97	**70.74**
$AA(\%)$	82.59	**85.04**	70.79	**71.27**
$kappa(\%)$	79.24	**81.30**	66.35	**68.26**
Class1	82.81	**83.10**	52.71	**53.27**
Class2	83.17	**84.96**	54.51	**62.03**
Class3	80.59	**84.36**	**90.89**	86.34
Class4	**90.81**	90.63	82.20	**83.62**
Class5	99.90	**100.00**	62.41	**72.63**
Class6	**95.80**	**95.80**	**81.82**	77.62
Class7	84.51	**91.70**	74.07	**78.73**
Class8	66.00	**67.52**	**81.48**	77.11
Class9	**85.55**	82.43	52.03	**63.64**
Class10	51.74	**58.11**	**72.97**	67.08
Class11	**66.89**	65.75	**83.78**	81.78
Class12	86.26	**88.28**	58.21	**61.10**
Class13	85.61	**88.42**	69.12	**75.44**
Class14	87.85	**95.14**	**70.85**	60.32
Class15	91.33	**99.37**	**74.84**	68.29

TABLE VI

COMPARISON OF SINGLE-MODAL RESULTS WITH AND WITHOUT DISTILLATION ON THE HS-SAR BERLIN DATASET. THE BOLD ITEMS DENOTE THE OPTIMAL VALUES IN THE ROWS

Methods	HS_Single	HS_DCM	SAR_Single	SAR_DCM
$OA(\%)$	64.74	**70.61**	46.85	**50.31**
$AA(\%)$	63.43	**64.20**	**26.16**	**26.16**
$kappa(\%)$	51.51	**57.69**	25.51	**27.74**
Class1	69.95	**77.78**	**29.46**	26.85
Class2	62.98	**71.95**	55.80	**62.85**
Class3	44.59	**50.22**	**46.40**	40.53
Class4	82.05	**83.02**	**68.67**	67.16
Class5	71.61	**76.85**	2.59	**5.59**
Class6	**69.42**	59.12	**5.54**	5.49
Class7	**38.22**	29.10	**0.80**	0.74
Class8	**68.59**	65.55	0.00	**0.08**

mine intermodal relationships, resulting in better classification performance than traditional algorithms. It is worth noting that the proposed method outperforms other compared methods by a large margin.

2) Comparison of Single-Modal Results: In this section, we introduce the scenario where we have both modalities in the training process while only one modality is available in the inference process. Table V reports the results of each single modality on the HS-LiDAR Housrton2013 dataset. HS_Single is a model trained by HS modality only. Similarly, LiDAR_Single is trained by LiDAR modality. It is worth noting that HS_single (LiDAR_single) has exactly the same structure as HS_DCM (LiDAR_DCM) during the inference stage. During the training stage, however, HS_DCM (LiDAR_DCM) will have an additional teacher model and DCM module.

We have observed that the proposed DCM module consistently enhances the performance of the single-modal model. By enabling the separation of the multimodal model from the single-modal model, the DCM module maintains equivalent computation and parameter levels to those of a single-modal model during inference. Furthermore, it facilitates the transfer of multimodal information to the single-modal model, resulting in improved classification outcomes. Notably, our DCM module consistently demonstrates performance enhancements on the HS-SAR Berlin dataset, as demonstrated in Table VI. In Section IV, we conduct a detailed analysis of the operational mechanism of the DCM. Due to the complementary nature of different modalities, the multimodal model can capture more cohesive class relationships. The DCM module enriches modality-specific models with multimodal models by injecting prior knowledge of class relationships.

E. Visualization and Analysis

1) Visualization of the Classification Maps in the Full Scene: To verify the effectiveness of the proposed distillation strategy from the visual perspective, we visualize the classification maps of different algorithms in the full scene with the HS-LiDAR Houston2013 and the HS-SAR Berlin testing set, respectively, which can be seen in Figs. 3 and 4.

1) As shown in Fig. 3(c), we can observe that the model with LiDAR modality only tends to be confused about soil and highway, while the model with only HS modality, as shown in Fig. 3(a) performs better. This may be because the LiDAR modality contains elevation information of the objects while the HS modality contains material information. These two categories, e.g., soil and highway, are more different from each other in terms of material information, resulting in better classification results with HS modalities. Since the multimodal model combines information from both modalities, both CCR-Net and our proposed model DIMNet show a better capability of classification than the single-modal model which can be seen in Fig. 3(e) and (f). In Fig. 3(d), it is worth noting that our proposed DCM module helps the model with LiDAR modality correct parts of the classification results, e.g., correcting parts of the misclassification results of soil.
2) Now, we mainly analyze the classification results of synthetic grass and highways. Different from soil and highway, the difference between synthetic grass and highway is more likely to lie in the elevation information. Therefore, we can observe that the model with HS modality only tends to be confused about synthetic grass and highway in Fig. 3(a), while the model with SAR modality only shows better performance in Fig. 3(c). Moreover, the multimodal model CCR-Net does not combine the information of SAR modality well, resulting in worse classification results. In contrast, our proposed multimodal model DIMNet correctly distinguishes these two categories. Similarly, our proposed DCM module helps the model with HS modality correct parts of the misclassification results of the highway.

2) Visualization of Internal Feature Representations: We give the visualization of the internal feature representations learned by various methods on both the HS-LiDAR Houston2013 and HS-SAR Berlin testing set. Results are

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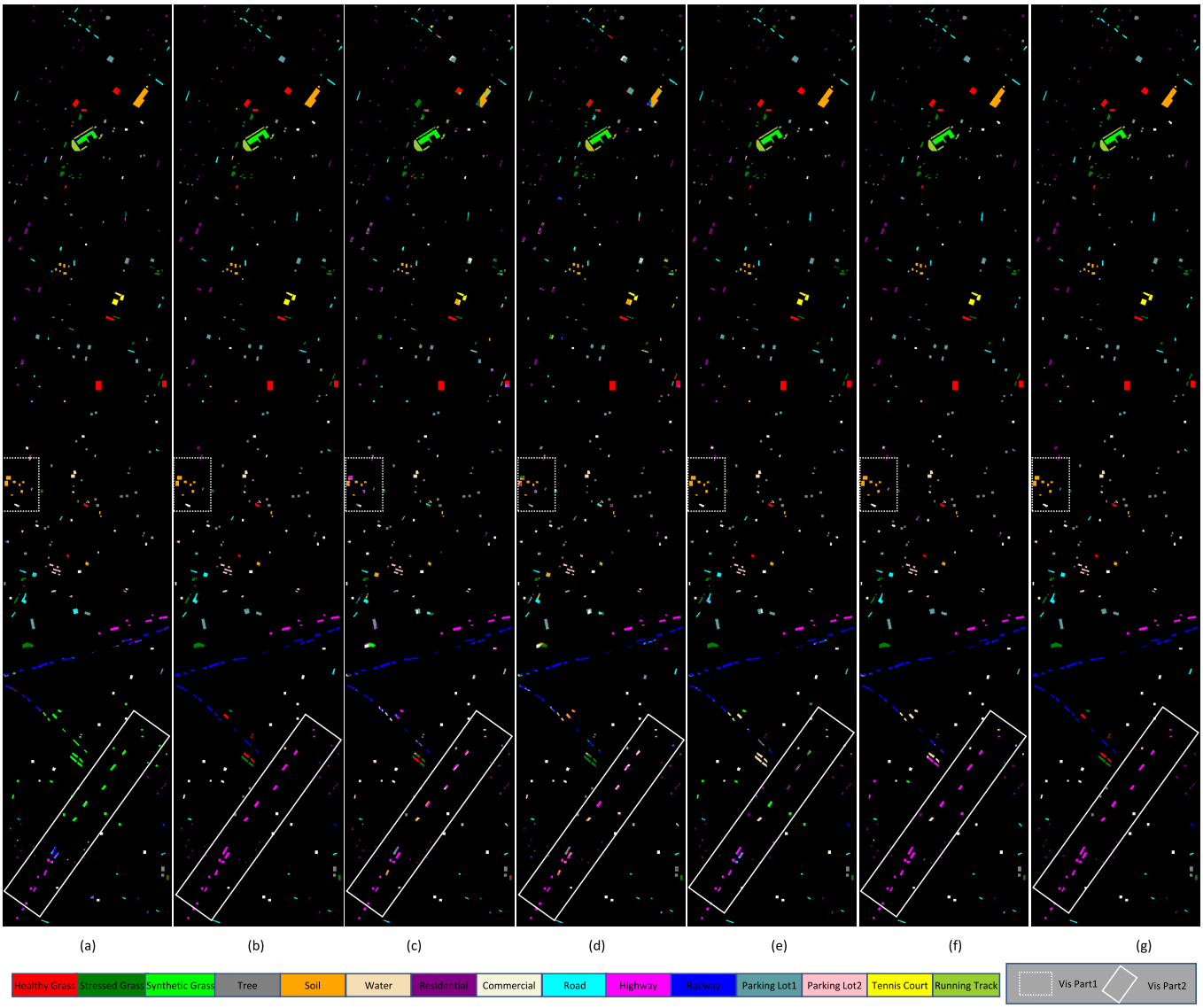


Fig. 3. Classification maps obtained by various methods with HS-LiDAR Houston2013 testing set. (a) HS_Single. (b) HS_DCM. (c) LiDAR_Single. (d) LiDAR_DCM. (e) CCR-Net. (f) DIMNet. (g) GT.

shown in Figs. 5 and 6, respectively. To this end, we randomly choose 140 samples per LC class in the testing set and their feature representations are extracted. Then, t-distributed stochastic neighbor embedding (t-SNE) [44] is applied to reduce the feature dimensionality to 2 for visualization purposes.

On both datasets, we can observe that LC class representations of multimodal models are more separable than those of single-modal models. LC class representations of the model with LiDAR (SAR) modality are less discriminative than that of the model with HS modality. These observations are consistent with the conclusions in Tables III and IV and Figs. 3 and 4. To analyze the influence of our proposed distillation strategy at the feature level, Fig. 5 is selected for a detailed explanation. By comparing Fig. 5(c) and (f), we can find that some LC class representations are still barely separable in CCR-Net model. Such ambiguities are successively alleviated by our proposed DIMNet model, especially for roads and highways. By comparing Fig. 5(a) and (b), our proposed DCM module helps the single-modal model with HS modality only to successfully separate the representations of road and parking lot1 apart, resulting better decision boundary. Moreover, in Fig. 5(d), the decision boundary between the tennis court and some LC class, e.g., soil, road, and parking lot1 is ambiguous. In Fig. 5(e), we can observe that our proposed DCM module allows the representation of the tennis court to be clearly separated from other categories.

F. Ablation Studies

In this section, we delve into a series of ablation experiments designed to evaluate the effectiveness and generalizability of the proposed ED, DIM, and DCM modules across different datasets and varying network architectures. First, we design ablation experiments to verify the effectiveness of the proposed ED and DIM modules on both the HS-LiDAR Houston2013 dataset and the HS-SAR Berlin dataset. Without the ED and DIM modules, our model reduces to a simple fusion

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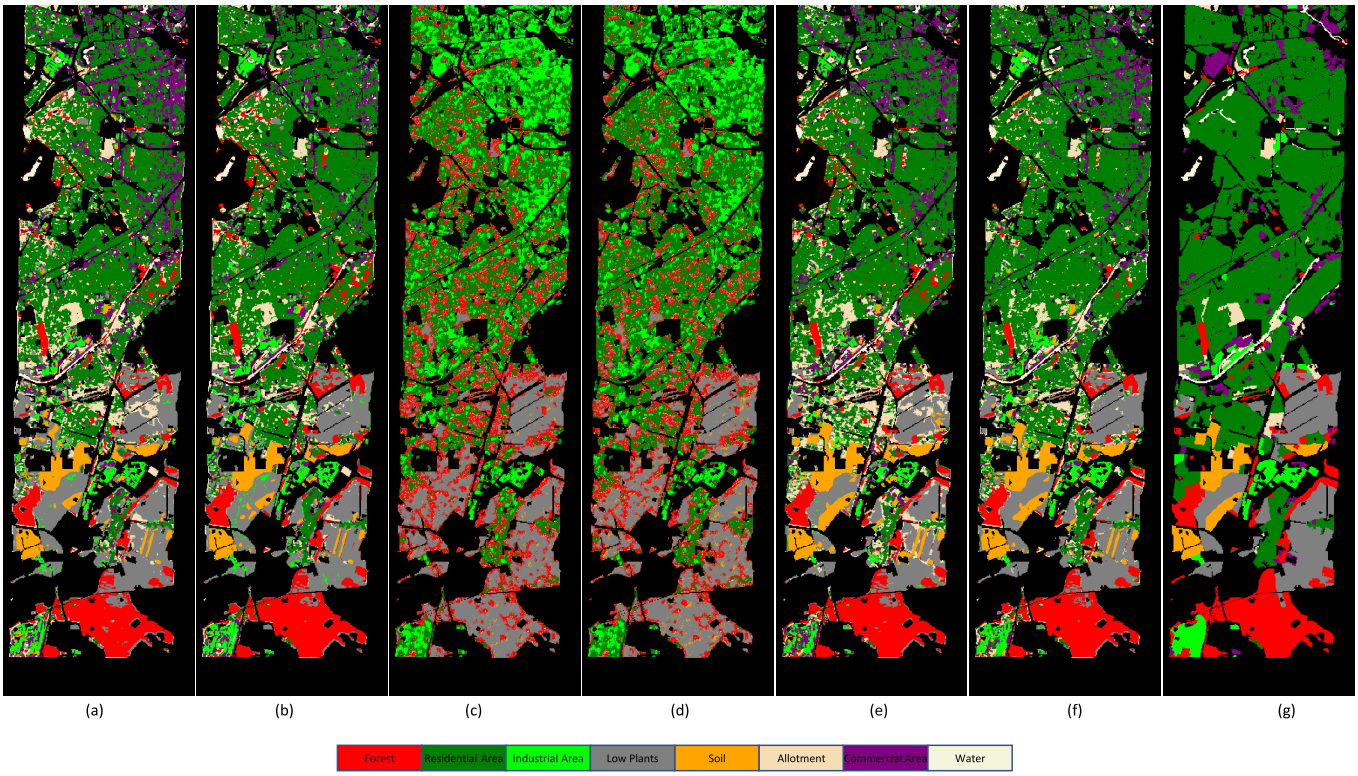


Fig. 4. Classification maps obtained by various methods on the HS-SAR Berlin testing set. (a) HS_Single. (b) HS_DCM. (c) SAR_Single. (d) SAR_DCM. (e) CCR-Net. (f) DIMNet. (g) GT.

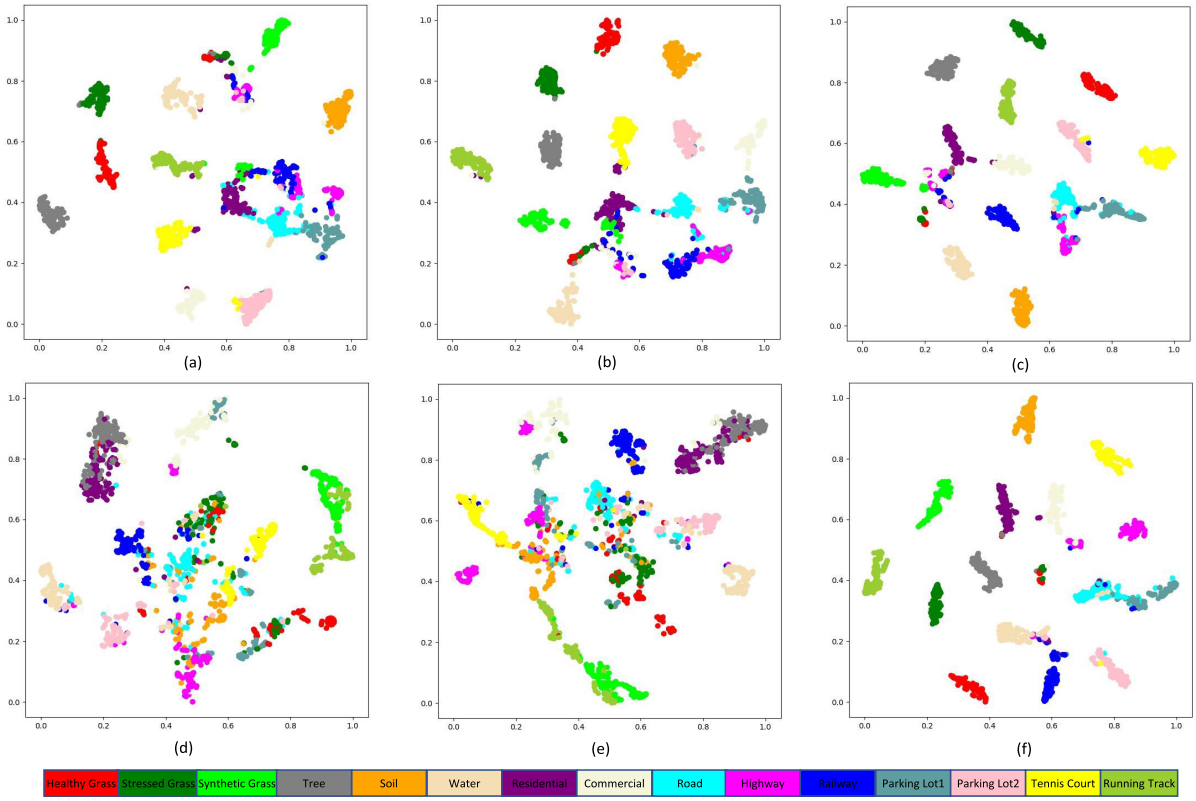


Fig. 5. t-SNE visualization of internal feature representation learned by various methods on the HS-LiDAR Houston2013 testing set. (a) HS_Single. (b) HS_DCM. (c) CCR-Net. (d) LiDAR_Single. (e) LiDAR_DCM. (f) DIMNet.

model (baseline model). This kind of fusion strategy directly concatenates the outputs of the two modalities and sends them to the classifier, which focuses on the decision level and ignores the information of weak modality, e.g., LiDAR. Then, to validate the good applicability of the proposed ED, DIM, and DCM modules, we replaced the feature extraction network

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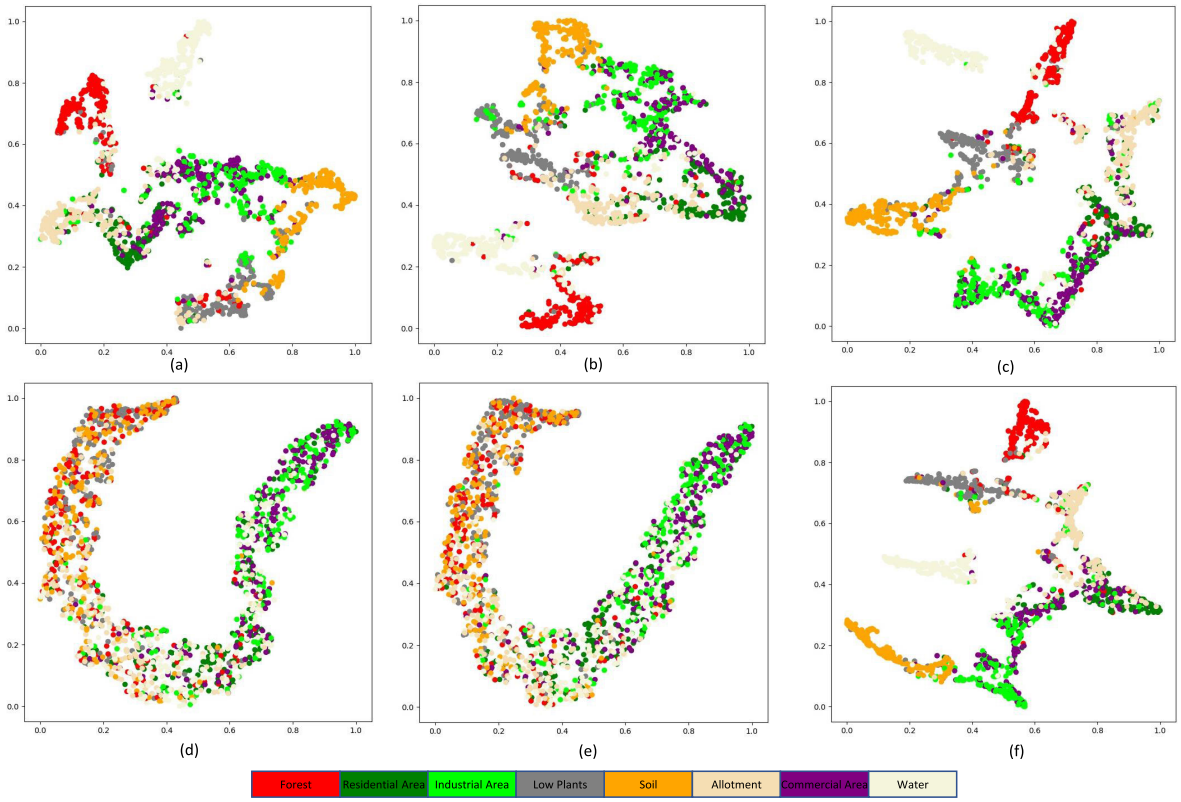


Fig. 6. t-SNE visualization of internal feature representation learned by various methods on the HS-SAR Berlin testing set. (a) HS_Single. (b) HS_DCM. (c) CCR-Net. (d) SAR_Single. (e) SAR_DCM. (f) DIMNet.

TABLE VII

ABLATION STUDIES OF THE PROPOSED ED AND DIM MODULES IN DIMNET ON THE HS-LIDAR HOUSTON2013 DATASET. THE BOLD ITEMS DENOTE THE OPTIMAL VALUES IN THE ROWS

Methods	baseline	ED	DIM	$ED\&DIM$
$OA(\%)$	88.27	90.96	90.74	**91.47**
$AA(\%)$	89.73	91.98	90.28	**92.48**
$kappa(\%)$	87.33	90.21	89.96	**90.77**
Class1	83.00	**83.10**	82.81	83.00
Class2	**85.15**	83.74	85.15	84.68
Class3	99.01	**99.21**	98.61	99.01
Class4	90.25	**92.52**	91.38	91.38
Class5	99.91	**100.00**	**100.00**	99.62
Class6	94.41	**95.10**	83.92	**95.10**
Class7	95.62	94.78	**95.80**	92.91
Class8	84.43	86.51	84.81	**87.27**
Class9	87.91	86.59	**91.50**	88.01
Class10	72.49	**93.82**	**93.82**	**93.82**
Class11	83.30	84.72	82.16	**88.80**
Class12	87.99	94.72	98.08	**96.54**
Class13	88.42	89.82	71.58	**90.53**
Class14	95.55	95.55	97.98	**96.76**
Class15	98.52	99.58	96.62	**99.79**

TABLE VIII

ABLATION STUDIES OF THE PROPOSED ED AND DIM MODULES IN DIMNET ON THE HS-SAR BERLIN DATASET. THE BOLD ITEMS DENOTE THE OPTIMAL VALUES IN THE ROWS

Methods	baseline	ED	DIM	$ED\&DIM$
$OA(\%)$	65.15	69.50	72.44	**73.73**
$AA(\%)$	61.63	63.04	64.28	**66.60**
$kappa(\%)$	51.44	55.52	59.37	**61.11**
Class1	73.89	76.57	**76.79**	74.46
Class2	63.23	72.84	74.99	**76.90**
Class3	42.37	**51.68**	49.29	48.56
Class4	81.49	68.78	**86.42**	**85.28**
Class5	75.65	**93.04**	92.27	**86.11**
Class6	52.64	39.32	26.54	**56.70**
Class7	**46.61**	32.54	27.83	26.60
Class8	57.18	69.55	**80.11**	**75.64**

of both the teacher model and the student model with blocks based on CNN and G2C [50], respectively.

As shown in Tables VII and VIII, we observe that the ED and DIM modules achieve consistent improvements against the baseline model. ED module obtains a compact representation by reconstruction via a self-supervised learning strategy. Though the constraint of reconstruction loss, it ensures that this compact representation can contain all the information of the two modalities, thus avoiding the representation dominated by information from one certain modality. By distilling information, the DIM module forces the output of each modality to mime the behavior of the fused modalities, which improves the representation ability of each modality and in turn promotes the effect of fusion. Specifically, the OA index from the Houston2013 dataset improves by 2.69% and 2.47% with the ED and DIM modules, respectively. For the Berlin dataset, the respective improvements are even more pronounced, with a 4.35% and 7.29% increase in OA. These results underline the effectiveness of each module in enhancing the model's performance. Furthermore, when both the ED and DIM modules

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TABLE IX

ABLATION STUDIES OF THE PROPOSED ED AND DIM MODULES IN DIMNET-G2C AND DIMNET-RNN ON THE HS-LIDAR HOUSTON2013 DATASET. THE BOLD ITEMS DENOTE THE OPTIMAL VALUES IN THE ROWS

DIMNet-G2C				
Methods	baseline	ED	DIM	$ED\&DIM$
$OA(\%)$	88.59	89.95	90.97	**92.08**
$AA(\%)$	89.76	91.05	91.94	**92.56**
$kappa(\%)$	87.74	89.19	90.28	**91.43**
DIMNet-RNN				
Methods	baseline	ED	DIM	$ED\&DIM$
$OA(\%)$	81.98	83.71	85.09	**86.68**
$AA(\%)$	84.52	85.83	87.12	**88.25**
$kappa(\%)$	75.63	77.21	78.49	**79.93**

are employed together, the performance of the models on both datasets is further enhanced, reaching optimal levels. The OA increases to 92.48% on the Houston2013 dataset and 73.73% on the Berlin dataset, surpassing the baseline performance by 4.21% and 8.58%, respectively. This demonstrates the complementary nature of the ED and DIM modules and their collective benefit in our proposed DIMNet model.

To further explore the adaptability of the proposed modules, we have replaced the CNN blocks in the feature extraction network of our DIMNet with G2C [50] blocks and RNN [51] blocks. The G2C block is inspired by the advancements in gradient-level feature extraction methods, integrating a dual convolution strategy that combines the strengths of traditional vanilla convolution and the novel gradient centralized convolution. More details can be found in [50] and the codes[3] are openly available. For the RNN, there are two recurrent layers with the gated recurrent unit (GRU). Each of them has 128 neuron units. The codes are openly available from https://github.com/danfenghong/HyFTech.

Table IX presents the results of ablation experiments on the HS-LiDAR Houston2013 dataset, which aim to evaluate the effectiveness of the proposed ED and DIM modules in the DIMNet-G2C and DIMNet-RNN model. In contrast to Table VII, we replaced the CNN blocks in the feature extraction network of DIMNet with G2C or RNN blocks, called DIMNet-G2C model or DIMNet-RNN model. Similar to the observations in the DIMNet model, the comparison results demonstrate the effectiveness of the ED and DIM modules within the DIMNet-G2C and DIMNet-RNN models, utilizing the HS-LiDAR Houston2013 dataset. The results underscore the contributions of each module to performance improvements, as evidenced by the increases in OA, AA, and kappa indices, reinforcing their efficacy in both G2C and RNN architectures.

Further validation of our DCM module was conducted by employing different feature extraction networks in both the teacher and student models, as depicted in Fig. 7. The student model configuration in Fig. 7(a) aligns with the setup detailed in Table V. We scrutinized three distinct teacher models—DIMNet-RNN, DIMNet, and DIMNet-G2C—and noted similar enhancements across key performance metrics,

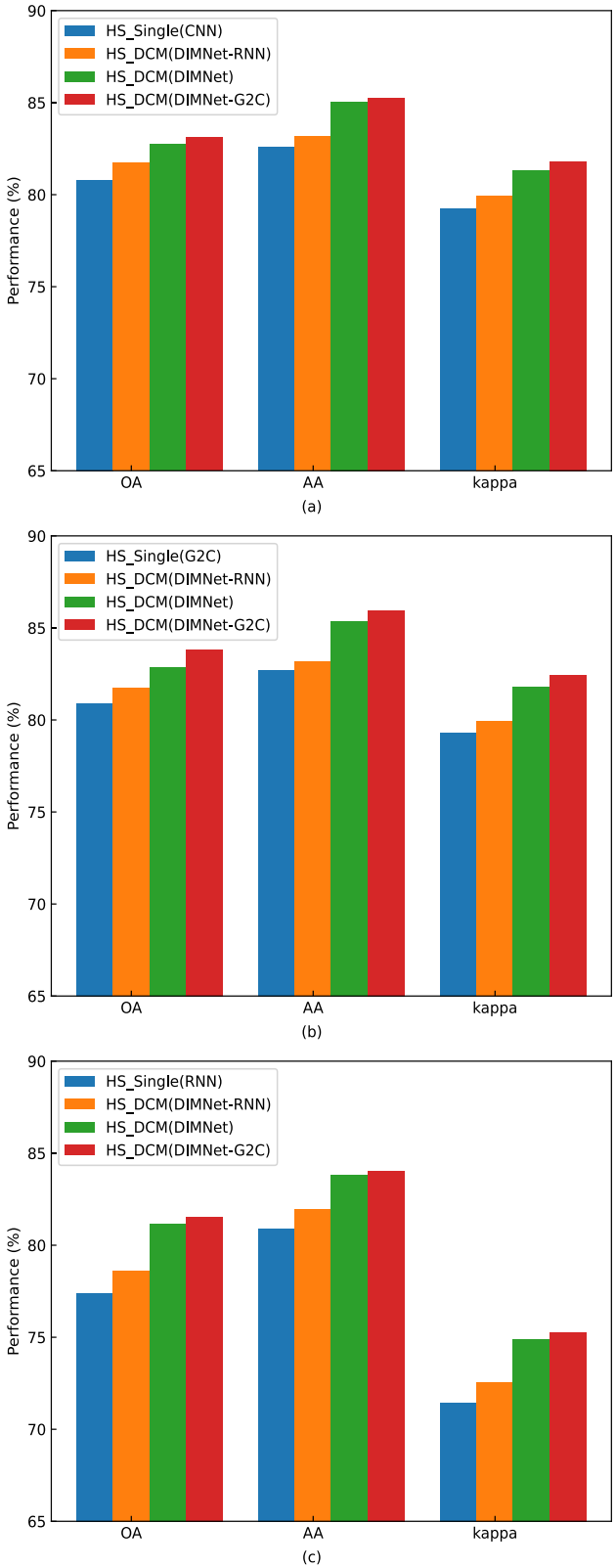


Fig. 7. Evaluation of DCM module generalization across diverse feature extractors in student and teacher networks. (a) DCM module ablation with CNN student model. (b) DCM module ablation with CNN student model. (c) DCM module ablation with CNN student model.

including OA, AA, and kappa. Fig. 7(b) and (c) shows the student model's feature extraction network, which was adapted to include G2C Blocks and RNN blocks, respectively. The

[3]https://github.com/researchgroup/G2C-Conv3D

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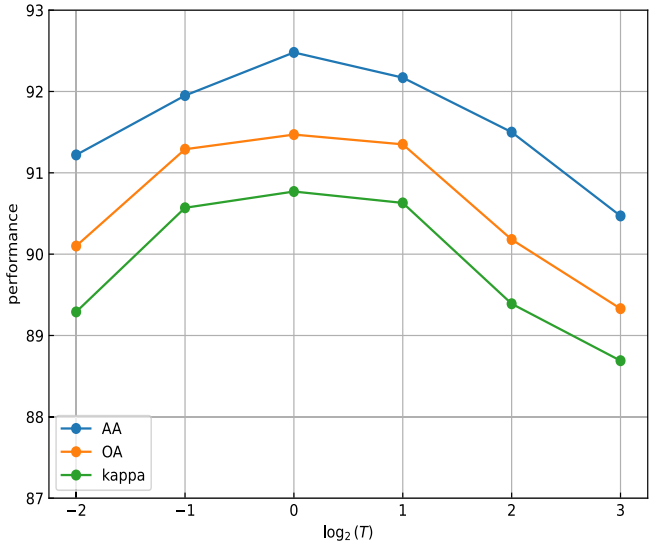


Fig. 8. Impact of temperature in the DIM module on the HS-LiDAR Houston2013 testing dataset. The temperature T is transformed into $\log_2(T)$ to maintain equal intervals.

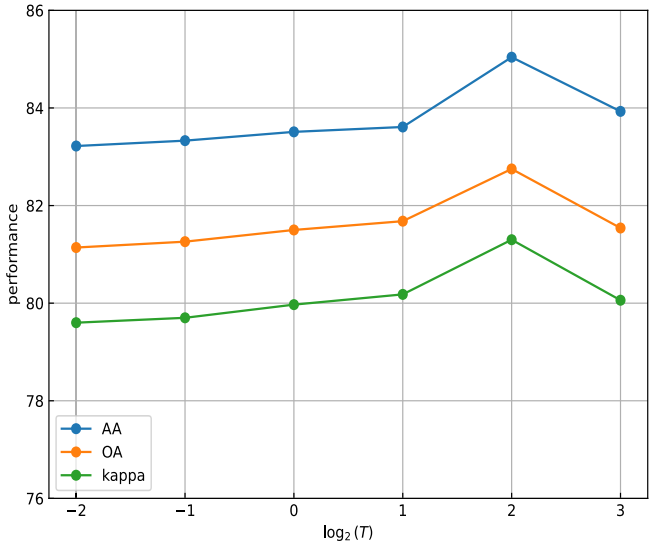


Fig. 9. Impact of temperature in the DCM module on the HS-LiDAR Houston2013 testing dataset with HS modality. The temperature T is transformed into $\log_2(T)$ to maintain equal intervals.

integration of the DCM module with these modifications consistently elevated the student model's performance. Collectively, these ablation studies verify the versatility of our proposed method, demonstrating its consistent applicability and effectiveness across a spectrum of feature extraction networks.

G. Parametric Studies

In this section, we provide a concise analysis of the hyperparameters T utilized in this article. The objective of this investigation is twofold: first, to ascertain the influence of these hyperparameters on the stability of the training process, and second, to evaluate the sensitivity of the model's performance with respect to these parameters.

Figs. 8 and 9 analyze the impact of the temperature T in the DIM and DCM modules, respectively. The experiments indicated that variations in the temperature T within both the DIM and DCM modules do influence the experimental outcomes, but the magnitude of these effects is minimal. Furthermore, these variations consistently improved the performance relative to the baseline. We first separately analyzed the impact of the temperature T in the DIM and DCM modules, as shown in Figs. 6 and 7. The experiments indicated that variations in the temperature T within both the DIM and DCM modules do influence the experimental outcomes, but the magnitude of these effects is minimal. Furthermore, these variations consistently improved the performance relative to the baseline.

In the DCM module, we found that setting the temperature T to 1 yielded optimal results. Conversely, in the DIM module, the best results were achieved when the temperature T was set to 4. It is noteworthy that on other datasets, different values might perform better. However, considering the minor impact of the temperature T and in the interest of simplicity, we empirically set the temperature T to 1 in all our DIM module experiments and to 4 in all our DCM module experiments. It is discerned that the optimal temperature T is 1 for the DCM module and 4 for the DIM module. Nevertheless, given the marginal influence exerted by the temperature T and in pursuit of methodological simplicity, we have taken an empirical approach. Consequently, in all our DIM module experiments, we have maintained a standard setting of T at 1, and in all our DCM module experiments, T has been consistently set at 4.

IV. Discussion

Our work primarily addresses two issues. First, the information in the fusion layer of the multimodal model is susceptible to being influenced by one modality, resulting in the network converging to a local optimal solution. Second, we investigate how to enhance the performance of single-modal models by leveraging multimodal data during the training phase. This section provides an in-depth analysis of the mechanisms behind our proposed modules from two perspectives: first, we provide new insights to explain how the multimodal model informs the class relationships to the single-modal model at optimality and improves model quality. Second, we investigate the impact of our proposed fusion approach on the fusion layer of the multimodal model.

A. Enhancing Modality-Specific Models With Multimodal Model by Injecting Class Relationship Prior Knowledge

Due to the complementarity among different modalities, multimodal models can capture more coherent class relationships. In this work, we found that the multimodal model's predictions on incorrect classes also provide a prior for single-modal model training. For instance, in the case of LiDAR modality, it is often challenging to distinguish between soil and highways. In this scenario, if a point represents soil, the predicted scores for these two classes using a single-modal

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model may be closer, such as 0.7 and 0.2. However, the multimodal model, having learned from the information provided by the HS modality, is able to effectively differentiate between these two categories. As a result, the predicted scores for these classes will be more distinct, for example, 0.9 and 0.01. When using one-hot labels, all negative classes receive equal penalties, making it difficult for the single-modal model to learn the relationship between soil and highways effectively. Next, we provide an optimization-based analysis to demonstrate how our proposed DCM module enables a single-modal model to learn such class relationship priors.

Let's consider the task of LC classification with K classes. Given a pair $(\boldsymbol{x}, \boldsymbol{y}) \in (X \times Y)$, where $\boldsymbol{x}$ represents the input and $\boldsymbol{y}$ is a one-hot encoded label ($\boldsymbol{y} \in \{0,1\}^K$), we aim to learn a parametric mapping function $f(\boldsymbol{x};\theta) : \boldsymbol{X} \to Y$. Here, $\theta \in \Theta$ refers to the parameters of the single-modal model. Our objective is to optimize these parameters θ using the Adam optimization algorithm

$$\begin{aligned}\theta^* &= \operatorname{argmin}_{\theta\in\Theta} L(\boldsymbol{y}, \boldsymbol{y}_s, f(\boldsymbol{x};\theta), \gamma, T) \\ &= H(\boldsymbol{y}, \boldsymbol{y}_s) + \gamma H(\hat{\boldsymbol{y}_{\text{fuse}}}, \hat{\boldsymbol{y}}_s)\end{aligned} \tag{16}$$

where L is the cross-entropy loss; $\boldsymbol{y}_s = f(\boldsymbol{x},\theta)$ is the network's output distribution computed by applying softmax over the logits $\boldsymbol{z}$; γ is a hyper-parameter; and $\hat{\boldsymbol{y}}_s$ and $\hat{\boldsymbol{y}}_{\text{fuse}}$ represent the temperature-softened predictions of the single-modal model (student model) and multimodal model (teacher model). The gradient of a single sample with respect to the logit z_i is given as follows:

$$\frac{\partial L}{\partial z_i} = \left(y_{s,i} - y_i\right) + \left(\frac{\gamma}{T}\right)\left(\hat{y}_{s,i} - \hat{y}_{\text{fuse},i}\right). \tag{17}$$

With the DCM module, the optimal solution of student's final logit layer weights $w_k^*, \forall k \in [K]$ enforces different inter-class distances based on teacher's probability distribution $\boldsymbol{y}_s$

$$||\boldsymbol{h} - \boldsymbol{w}_i^*||^2 < ||\boldsymbol{h} - \boldsymbol{w}_j^*||^2, \quad \text{iff } p_i > p_j \quad \forall p_i > p_j \quad i, j \in [K] \tag{18}$$

where $\boldsymbol{h}$ is the activations of the penultimate layer.

Proof: To find the optimal solution for the student model, we set the gradient in (17) equal to zero

$$y_{s,k}^* + \frac{\gamma}{T}\hat{y}_{s,k}^* = y_k + \frac{\gamma}{T}\hat{y}_{\text{fuse},k}. \tag{19}$$

Using a similar proof technique as Müller et al. [45], $||\boldsymbol{h} - \boldsymbol{w}_k^*||^2 = ||h||^2 + ||w_k^*||^2 - 2\boldsymbol{h}^{\mathrm{T}} w_k^*$, where $\boldsymbol{h}$ is the penultimate layer activations, and w_k^* are the weights of the last logits layer for class $k \in [K]$. Note that $||h||^2$ is factored out when computing the softmax, and $||w_k^*||^2$ is usually a (regularized) constant across all classes. Equating $z_k^* = \boldsymbol{h}^{\mathrm{T}} w_k^*$, and using the property $\text{softmax}(z) = \text{softmax}(z + c)$, $\forall c \in \mathbb{R}$, we get

$$\begin{aligned} y_{s,k}^* &= \text{softmax}\left(z_k^*\right) \\ &= \text{softmax}\left(\boldsymbol{h}^{\mathrm{T}} \boldsymbol{w}_k^*\right) \\ &= \text{softmax}\left(-\frac{1}{2}||\boldsymbol{h} - \boldsymbol{w}_k^*||^2\right). \end{aligned} \tag{20}$$

By substituting the derived expression into (19), we obtain

$$\text{softmax}\left(-\frac{1}{2}||\boldsymbol{h} - \boldsymbol{w}_k^*||^2\right) + \frac{\gamma}{T}\text{softmax}\left(-\frac{1}{2T}||\boldsymbol{h} - \boldsymbol{w}_k^*||^2\right) = y_k + \frac{\gamma}{T}\hat{y}_{\text{fuse},k}. \tag{21}$$

Note that softmax is a monotonically increasing function. Thus, we can express the left-hand side (LHS) as follows:

$$g\left(-\frac{1}{2}||\boldsymbol{h} - \boldsymbol{w}_k^*||^2; \gamma, T\right) = y_k + \frac{\gamma}{T}\hat{y}_{\text{fuse},k} \tag{22}$$

where $g(x;\gamma,T) : \mathbb{R} \to \mathbb{R}$ is a monotonically increasing function, parameterized by γ and T. When considering the incorrect classes where $y_k = 0$, it is important to observe that softmax temperature scaling maintains the relative ordering of the teacher's probabilities $\hat{y}_{\text{fuse},k}$. This observation helps us establish the validity of the claim.

In addition, the visualization results further support our hypothesis. Taking the example of a single-modal model using LiDAR data, as illustrated in Fig. 3(c), we observe that the model with only LiDAR data tends to confuse soil and highway classes. However, the multimodal model exhibits better discrimination between these two classes. When we inject the prior distribution of the multimodal model on these classes into the single-modal model using the DCM module, we can see in Fig. 3(d) that our proposed DCM module helps correct misclassifications in the LiDAR modality, such as rectifying misclassifications of soil.

B. Avoiding Dominance of Modality-Specific Information in Fusion Layers

The richness of information contained in different modalities varies significantly, and existing models have largely overlooked this aspect when fusing multimodal information. Due to the nonconvexity of neural networks, we typically only achieve local optimal solutions during the training process. If the neural network finds a shortcut during training, it can easily fall into a suboptimal local solution. Optical RS images typically contain more information, so during the training process, the fusion layer tends to utilize information from optical RS images. This allows the training loss to decrease faster, leading to higher activation of neurons from the optical RS images. However, once trapped in such a local optimal solution, information from the other modality is utilized less.

To mitigate this problem, we designed the ED and DIM modules to ensure better extraction of different modality information during the fusion process. The ED module forces the fused layer's information to be able to reconstruct the different modality information before fusion, which can ensure to a certain extent that the fusion layer includes more information from both modalities. Meanwhile, the DIM module allows the classification logits of different modalities to mimic the fused logits, further enabling each branch to extract more classification information.

Fig. 1 presents a comparative analysis of the activation and weightage of neurons in the fusion layer across different methods. Fig. 6(a) and (b) depicts that the absence of both DIM and ED modules leads to a higher contribution from

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the HS modality compared to the LiDAR modality. This dominance is significantly balanced upon the introduction of the ED module, and even more so with the addition of the DIM module.

Specifically, Fig. 1(a) shows that without the DIM and ED modules, the number of activated neurons from the HS modality in the fusion layer is 36.8% more than those from the LiDAR modality. This dominance reduces to 26.0% with the inclusion of the ED module and further to 17.0% with the full-fledged DIMNet model.

Similarly, Fig. 1(b) shows that in terms of the sum of neuron weights, the absence of both modules leads to a 21.6% higher contribution from the HS modality. This discrepancy reduces to 13.6% with the ED module and further to 7.8% with the DIMNet model.

This clearly demonstrates the effectiveness of the DIM and ED modules in harmonizing the fusion process by equally leveraging the information from both the HS and LiDAR modalities. This balanced interplay facilitates the optimal utilization of the multimodal data, consequently enhancing the overall performance of the model.

Visualization of the predicted results further verifies the effectiveness of our proposed approach, as shown in Fig. 3. In the task of distinguishing synthetic grass and highways, the classification result of the single model of the LiDAR modality is clearly superior to that of the HS modality. However, CCR-Net did not successfully integrate SAR modality information, resulting in unsatisfactory classification results, while our DIMNet performed better in distinguishing these two categories. The same conclusion is drawn from the experimental results indicators. This demonstrates that our method indeed extracts different modalities' information more effectively.

V. Conclusion

In this article, we present a novel and robust LC classification framework incorporating multimodal knowledge distillation. To enhance the fusion of multimodal RS data, we have developed DIMNet, which integrates ED and DIM modules. ED module can fuse multimodal features in a more compact way and the DIM module enables the network to effectively transfer knowledge from the fused prediction to the per-source prediction. In addition, in scenarios where one modality is available during the inference stage while both are available during training, we have proposed the DCM module to enhance the performance of single-modal models. This module leverages multimodal information to the single-modal models. Experimental validation on the HS-LiDAR Houston2013 and HS-SAR Berlin verification sets confirms the effectiveness of our proposed framework. While our distillation strategies offer new insights into multimodal LC classification tasks, further exploration of their applicability with more than two modalities is warranted.

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References

[1] J. Chanussot, J. A. Benediktsson, and M. Fauvel, "Classification of remote sensing images from urban areas using a fuzzy possibilistic model," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 1, pp. 40–44, Jan. 2006.

[2] E. Bartholomé and A. S. Belward, "GLC2000: A new approach to global land cover mapping from Earth observation data," *Int. J. Remote Sens.*, vol. 26, no. 9, pp. 1959–1977, Feb. 2007.

[3] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[4] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[5] M. Berger, J. Moreno, J. A. Johannessen, P. F. Levelt, and R. F. Hanssen, "ESA's sentinel missions in support of Earth system science," *Remote Sens. Environ.*, vol. 120, pp. 84–90, 2012.

[6] M. Schmitt and X. X. Zhu, "Data fusion and remote sensing: An ever-growing relationship," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 4, pp. 6–23, Dec. 2016.

[7] P. Ghamisi et al., "Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 6–39, Mar. 2019.

[8] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[9] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[10] Y. J. E. Gbodjo, O. Montet, D. Ienco, R. Gaetano, and S. Dupuy, "Multisensor land cover classification with sparsely annotated data based on convolutional neural networks and self-distillation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11485–11499, 2021.

[11] W. Liu and R. Qin, "Unsupervised transfer learning using for multi-model remote sensing data classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 636–639.

[12] G. Li, C. Zhang, M. Wang, F. Gao, and X. Zhang, "Scene classification of high-resolution remote sensing image using transfer learning with multi-model feature extraction framework," in *Proc. IGTA*, Beijing, China, Apr. 2018, pp. 238–251.

[13] R. Kou, B. Fang, G. Chen, and L. Wang, "Progressive domain adaptation for change detection using season-varying remote sensing images," *Remote Sens.*, vol. 12, no. 22, p. 3815, Nov. 2020.

[14] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[15] W. Liao, A. Pižurica, R. Bellens, S. Gautama, and W. Philips, "Generalized graph-based fusion of hyperspectral and LiDAR data using morphological features," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 552–556, Mar. 2015.

[16] P. Ghamisi, J. A. Benediktsson, and S. Phinn, "Land-cover classification using both hyperspectral and LiDAR data," *Int. J. Image Data Fusion*, vol. 6, no. 3, pp. 189–215, Jul. 2015.

[17] J. Xia, N. Yokoya, and A. Iwasaki, "Fusion of hyperspectral and LiDAR data with a novel ensemble classifier," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 6, pp. 957–961, Jun. 2018.

[18] G. Camps-Valls, L. Gomez-Chova, J. Munoz-Mari, J. L. Rojo-Alvarez, and M. Martinez-Ramon, "Kernel-based framework for multitemporal and multisource remote sensing data classification and change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 6, pp. 1822–1835, Jun. 2008.

[19] L. Yan, M. Cui, and S. Prasad, "Joint Euclidean and angular distance-based embeddings for multisource image analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 7, pp. 1110–1114, Jul. 2018.

[20] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "CoSpace: Common subspace learning from hyperspectral-multispectral correspondences," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4349–4359, Jul. 2019.

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[21] D. Hong, N. Yokoya, N. Ge, J. Chanussot, and X. X. Zhu, "Learnable manifold alignment (LeMA): A semi-supervised cross-modality learning framework for land cover and land use classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 147, pp. 193–205, Jan. 2019.

[22] D. Feng et al., "Deep multi-modal object detection and semantic segmentation for autonomous driving: Datasets, methods, and challenges," 2019, *arXiv:1902.07830*.

[23] Y. Xu et al., "Advanced multi-sensor optical remote sensing for urban land use and land cover classification: Outcome of the 2018 IEEE GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1709–1724, Jun. 2019.

[24] Y. Chen, C. Li, P. Ghamisi, X. Jia, and Y. Gu, "Deep fusion of remote sensing data for accurate classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1253–1257, Aug. 2017.

[25] R. Hang, Q. Liu, D. Hong, and P. Ghamisi, "Cascaded recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5384–5394, Aug. 2019.

[26] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[27] D. Ienco, R. Gaetano, C. Dupaquier, and P. Maurel, "Land cover classification via multitemporal spatial data by deep recurrent neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1685–1689, Oct. 2017.

[28] K. K. Gadiraju, B. Ramachandra, Z. Chen, and R. R. Vatsavai, "Multimodal deep learning based crop classification using multispectral and multitemporal satellite imagery," in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2020, pp. 3234–3242.

[29] B. Lucas, C. Pelletier, D. Schmidt, G. I. Webb, and F. Petitjean, "Unsupervised domain adaptation techniques for classification of satellite image time series," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 1074–1077.

[30] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[31] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.

[32] X. Jiao et al., "TinyBERT: Distilling BERT for natural language understanding," 2019, *arXiv:1909.10351*.

[33] N. S. Keskar, D. Mudigere, J. Nocedal, M. Smelyanskiy, and P. T. P. Tang, "On large-batch training for deep learning: Generalization gap and sharp minima," 2016, *arXiv:1609.04836*.

[34] H.-C. Li, W.-Y. Wang, L. Pan, W. Li, Q. Du, and R. Tao, "Robust capsule network based on maximum correntropy criterion for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 738–751, 2020.

[35] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015, *arXiv:1502.03167*.

[36] J. Ngiam, A. Khosla, M. Kim, J. Nam, H. Lee, and A. Ng, "Multimodal deep learning," in *Proc. ICML*, 2011, pp. 689–696.

[37] J. Gou, B. Yu, S. J. Maybank, and D. Tao, "Knowledge distillation: A survey," *Int. J. Comput. Vis.*, vol. 129, no. 6, pp. 1789–1819, Jun. 2021.

[38] J. Yim, D. Joo, J. Bae, and J. Kim, "A gift from knowledge distillation: Fast optimization, network minimization and transfer learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 7130–7138.

[39] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, "How transferable are features in deep neural networks," in *Proc. Int. Conf. Adv. Neural Inf. Process. Syst.*, Montreal, QC, Canada, Dec. 2014, pp. 3320–3328.

[40] L. Beyer, X. Zhai, A. Royer, L. Markeeva, R. Anil, and A. Kolesnikov, "Knowledge distillation: A good teacher is patient and consistent," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10915–10924.

[41] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.

[42] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[43] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1026–1034.

[44] L. van der Maaten, "Accelerating t-SNE using tree-based algorithms," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 3221–3245, Oct. 2014.

[45] R. Müller, S. Kornblith, and G. E. Hinton," When does label smoothing help," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–13.

[46] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[47] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[48] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213.

[49] Y. Zhang et al., "Multimodal transformer network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514317.

[50] S. K. Roy, P. Kar, D. Hong, X. Wu, A. Plaza, and J. Chanussot, "Revisiting deep hyperspectral feature extraction networks via gradient centralized convolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516619.

[51] Q. Li, B. Zheng, B. Tu, J. Wang, and C. Zhou, "Ensemble EMD-based spectral–spatial feature extraction for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5134–5148, 2020.



Guozheng Xu received the B.E. degree in civil engineering from the School of Civil and Architectural Engineering, Beijing Jiaotong University, Beijing, China, in 2017, and the M.S. degree in architecture and civil engineering from the School of Naval Architecture, Ocean and Civil Engineering, Shanghai Jiao Tong University, Shanghai, China, in 2020, where he is currently pursuing the Ph.D. degree in information and communication engineering with the School of Electronic Information and Electrical Engineering.

His research interests include image processing and analysis, deep learning, and multimodal machine learning.

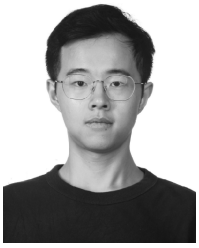


Xue Jiang (Senior Member, IEEE) received the M.S. degree in acoustics and signal processing from the Chinese Academy of Sciences, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from McMaster University, Hamilton, ON, Canada, in 2014.

From 2007 to 2010, she was a Research Engineer with Sunplus Inc., Beijing. She held a Post-Doctoral Fellow position at McMaster University from 2014 to 2015 and Arizona State University, Tempe, AZ, USA, from 2015 to 2016. She joined the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China, in 2016, where she is currently a Professor. Her research interests include array processing, robust beamforming, sparse and low-rank learning, data science, and numerical optimization, with applications to radar, sonar, and remote sensing.

Dr. Jiang was one of the Finalists of the Best Student Paper Award of the IEEE 2013 Radar Conference. She received the McMaster University International Excellence Awards in 2012 and 2013. She is an Associate Editor of IEEE TRANSACTIONS ON SIGNAL PROCESSING. She serves as the Editorial Board Member for *IET Radar, Sonar & Navigation*.

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Yue Zhou (Student Member, IEEE) received the B.S. degree in electronic and information engineering from the Beijing University of Posts and Telecommunications, Beijing, China, in 2017. He is currently pursuing the Eng.D. degree with the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China.

His research interests include deep learning, image processing, and radar object detection and recognition.



Shutao Li (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from Hunan University, Changsha, China, in 1995, 1997, and 2001, respectively.

In 2001, he joined the College of Electrical and Information Engineering, Hunan University, where he is currently a Full Professor. In 2001, he was a Research Associate with the Department of Computer Science, The Hong Kong University of Science and Technology, Hong Kong, where he was a Visiting Professor, in 2005. From 2002 to 2003, he was a Post-Doctoral Fellow with the Royal Holloway College, University of London, London, U.K. He has authored or coauthored more than 200 refereed articles. His research interests include image processing, pattern recognition, and artificial intelligence.

Dr. Li was a recipient of the two Second-Grade State Scientific and Technological Progress Awards of China in 2004 and 2006. He is a Member of the Editorial Board of *Information Fusion* and *Sensing and Imaging*. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT.



Xingzhao Liu (Member, IEEE) received the B.S. and M.S. degrees from the Harbin Institute of Technology, Harbin, China, in 1984 and 1992, respectively, and the Ph.D. degree from Tokushima University, Tokushima, Japan, in 1995, all in electrical engineering.

From 1984 to 1998, he was an Assistant Professor, an Associate Professor, and a Professor with the Harbin Institute of Technology. Since 1998, he has been a Professor with Shanghai Jiao Tong University, Shanghai, China. His research interests include high-frequency radar and synthetic aperture radar signal processing.



Peiwen Lin received the B.S. degree in physics and information engineering from Fuzhou University, Fuzhou, China, in 2016, and the M.S. degree in software engineering from Tianjin University, Tianjin, China in 2019.

His research interests include various aspects of artificial intelligence in the field of autonomous driving, including image classification, image segmentation, panoramic segmentation, and multimodal learning.

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